CS & IT ENGINEERING

Programming in C

Functions and Storage Classes

Lec-04

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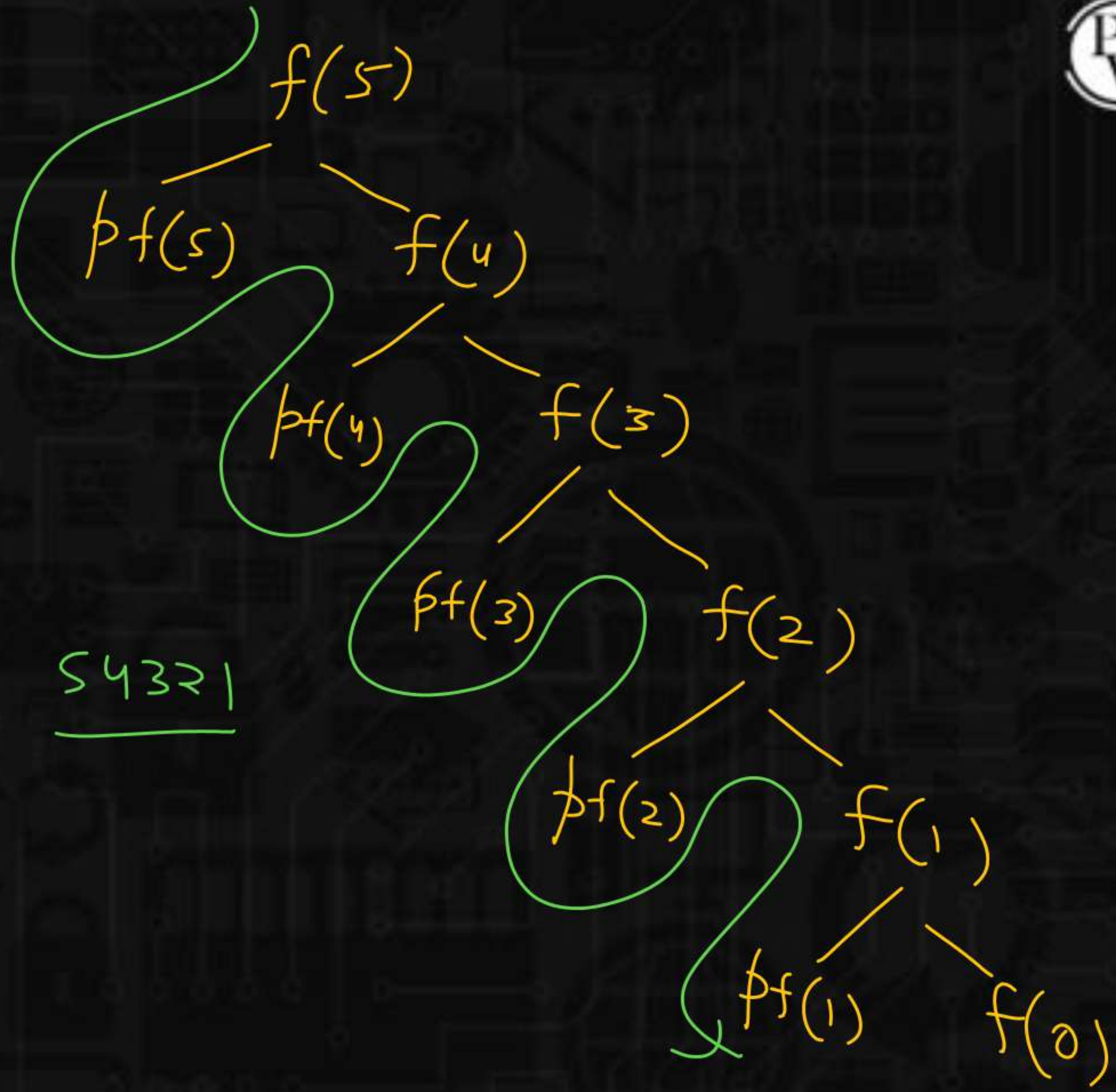
TOPICS TO BE COVERED

Recursion

Q.1

```
void f(int n)
   {
      if(n<=0)
      return;
      printf("%d",n);
      f(n-1);
   }
```

What is the output of f(5)



54321

Q.2

```
void f(int n)
    {
        if(n<=0)
        return;
        f(n-1);
        printf("%d",n);
    }
```

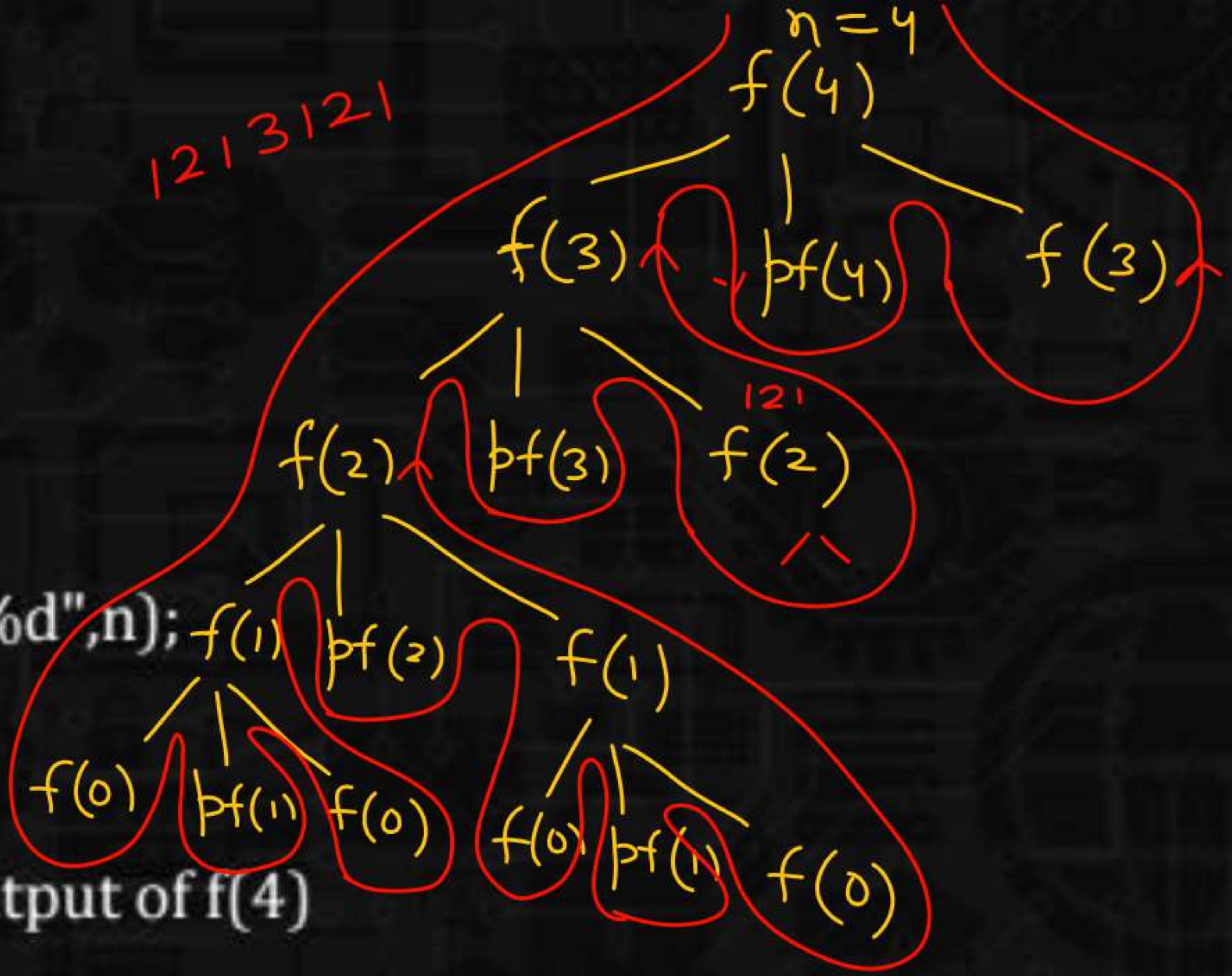
What is the output of f(5)

12345

Q.3

```
void f(int n)
{
    if(n<=0)
    return;
  1 f(n-1);
  2 printf("%d",n);
  3 f(n-1);
}
```

What is the output of f(4)

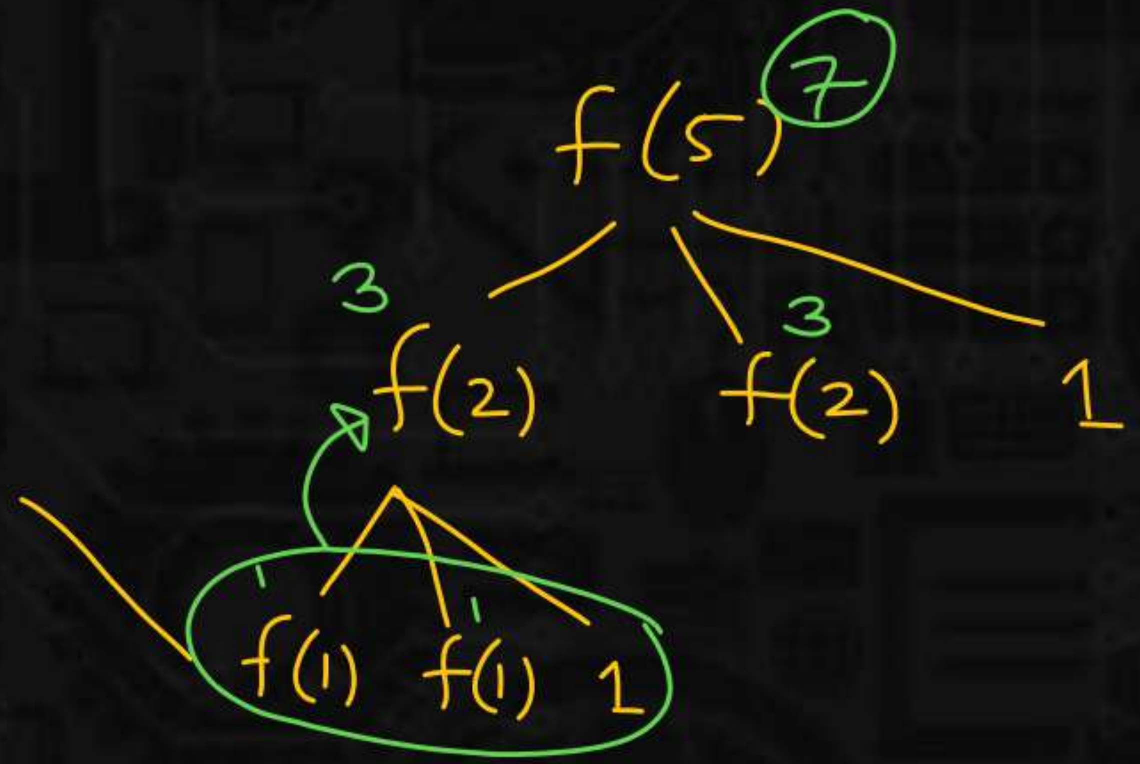


1213121 4 1213121

Q.4

```
int f(int n)
{
    if(n<=1)
    return n;
    return f(n/2) + f(n/2) + 1;
}
```

What is the output of f(5)



Q.5

```
int f(int n)
    {
        if(n<=1)
        return n;
        return f(n/2) + n/2 ;
    }
```

What is the output of f(12)

$n = 12$

$f(12)$ 11

$f(6)$ 5 $+ 6$

$f(3)$ 2 $+ 3$

$f(1)$ 1 $+ 1$

Q.6

int f(int n)

{

if(n<=1)

return n;

if(n%2)

return f(n/2) + n; odd

return f(n/3) + n; Even

}

output of f(22) ?

$n = 22$

$\frac{22}{3} = 7$

$f(22) \leftarrow 33$

$n = 7$ 11

$f(7) + 22$

4

$f(3) + 7$

1

$f(1) + 3$

1

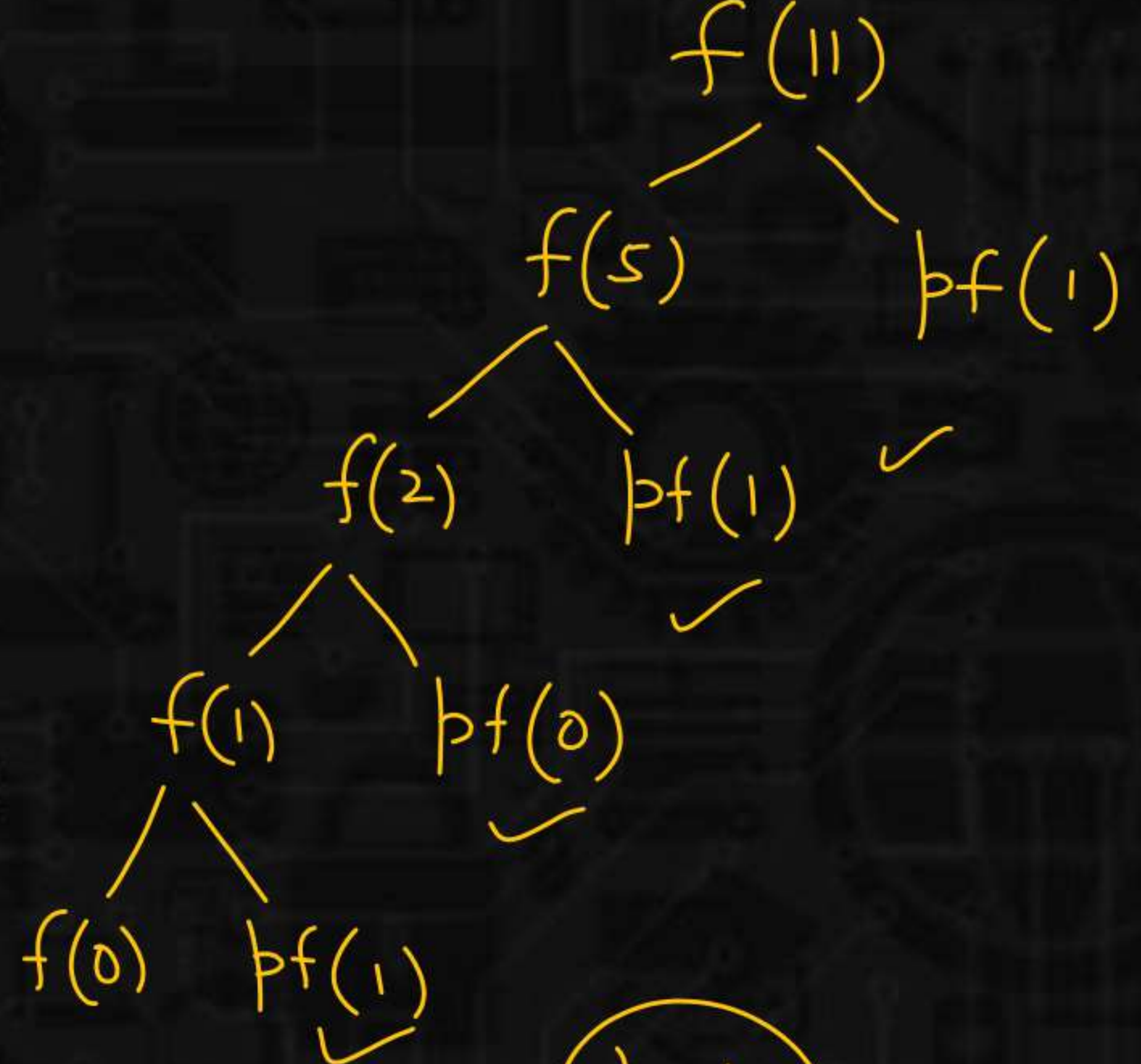
```
fun( ){

    if (n<=1)        true
      return n;

    return n+2;
    }

}
```

Q.7

Consider the code :

```
/* Assume that n>=0 */
void fun(int n)
 {
     if(n==0)
     return ;
 (1) fun(n/2);
 (2) printf("%d",n%2);
 }
```

output of f(11)?



1011

3min

Q.8

Consider the following C program :

```
void foo(int n , int sum) {
    int k=0,j=0;
    if(n==0)
    return;
    k=n%10;
    j=n/10;
    sum=sum + k;
    foo(j,sum);
    printf("%d",k);
}
void main(){
1.  int a=2048,sum=0;
2.  foo(a,sum);
3.  printf("%d",sum);
}
```

Output?

~~A.~~ 8, 4, 0, 2, 14

B. 8,4,0,2,0

~~C.~~ 2,0,4,8,14

D. 2,0,4,8,0

Q.8

3min

Consider the following C program :

```
void foo(int n , int sum) {
    int k=0,j=0;
    if(n==0)
    return;
    k=n%10;
    j=n/10;
    sum=sum + k;
    foo(j,sum);
    printf("%d",k);
}
void main(){
int a=2018,sum=0;
foo(a,sum);
printf("%d",sum);
}
```

Output?

- ~~A.~~ 8, 4, 0, 2, 14
- B. 8,4,0,2,0
- ~~C.~~ 2,0,4,8,14
- D. 2,0,4,8,0 ✓

K = 8
j = 204
sum = 8
foo(204,8)
⇒ rec. call

8
4
0
2

foo(2048,0)

2,0,4,8,0

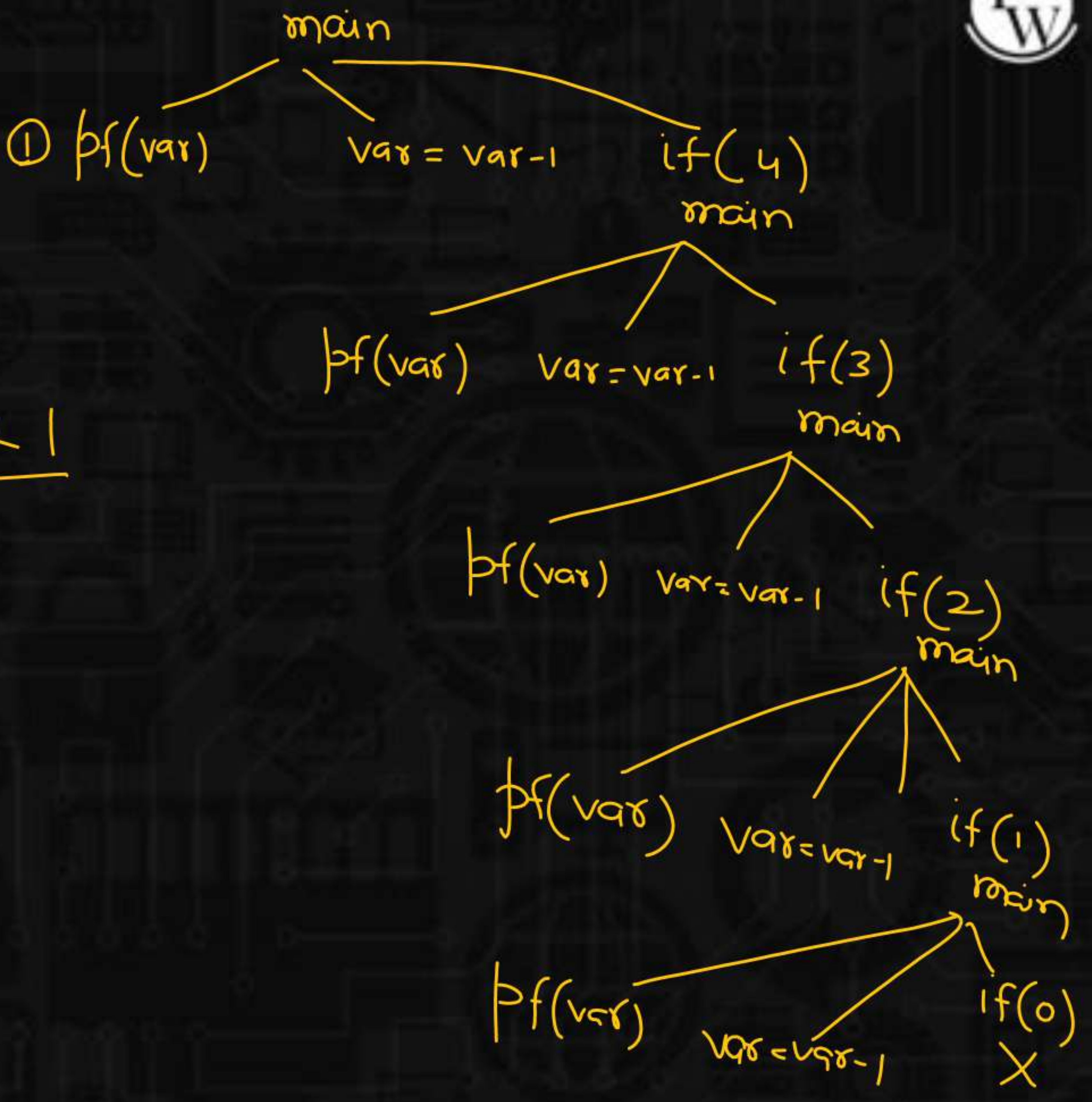
Q.9

```c
void main()
{
    static int var=5;
    printf("%d",var--);
    if(var)
    main();
}
```

data seg

var $\boxed{\cancel{5}\,\cancel{4}\,\cancel{3}\,\cancel{2}\,\cancel{1}\,0}$

5 4 3 2 1



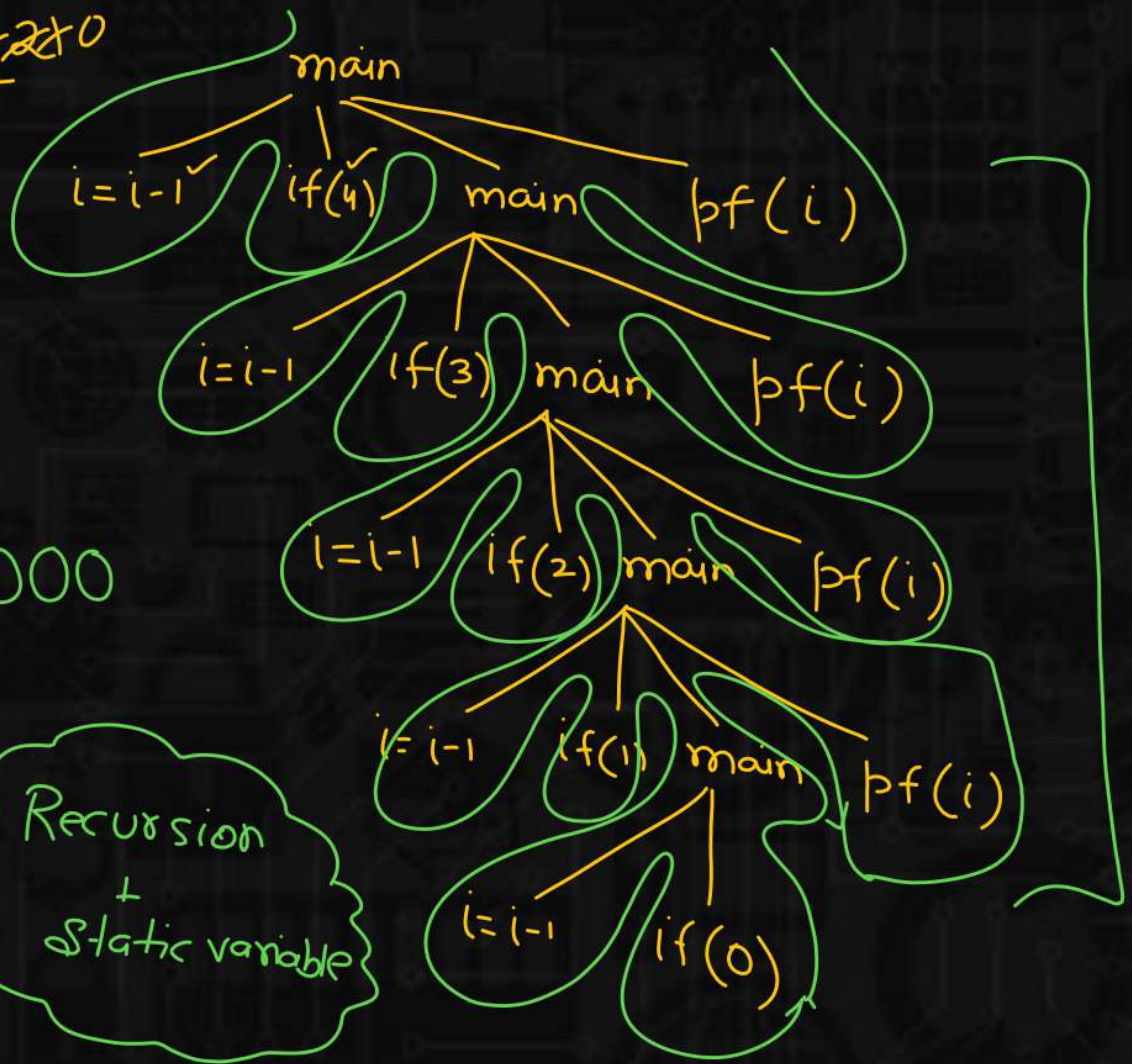
Q.10

```
void main()
{
    static int i =5;
    if(--i)
    {
      1. main();
      2. printf("%d",i);
    }
}
```

i = i-1   if(i)

i $\boxed{\cancel{5}\,\cancel{4}\,\cancel{3}\,\cancel{2}\,\cancel{1}\,0}$

0000



Respect : Recursion + Static variable

Q.11

predict the output

```
int fun(int x)
  {
      if(x%2==0)
      return fun(fun(x-1));
    else
      return(x++);
  }
int main()
  {
                 fun
      printf("%d",f(12));
      [redacted]
      return 0;
  }
```

return x;
x=x+1;

(A.) 10

(B.) 11

(C.) 12

(D.) None of these

x=11
fun(11)

x=12
fun(12)
|
fun(fun(11))
↓
fun(11)
↓
11

fun(11) $x = 11$

Q.11

predict the output

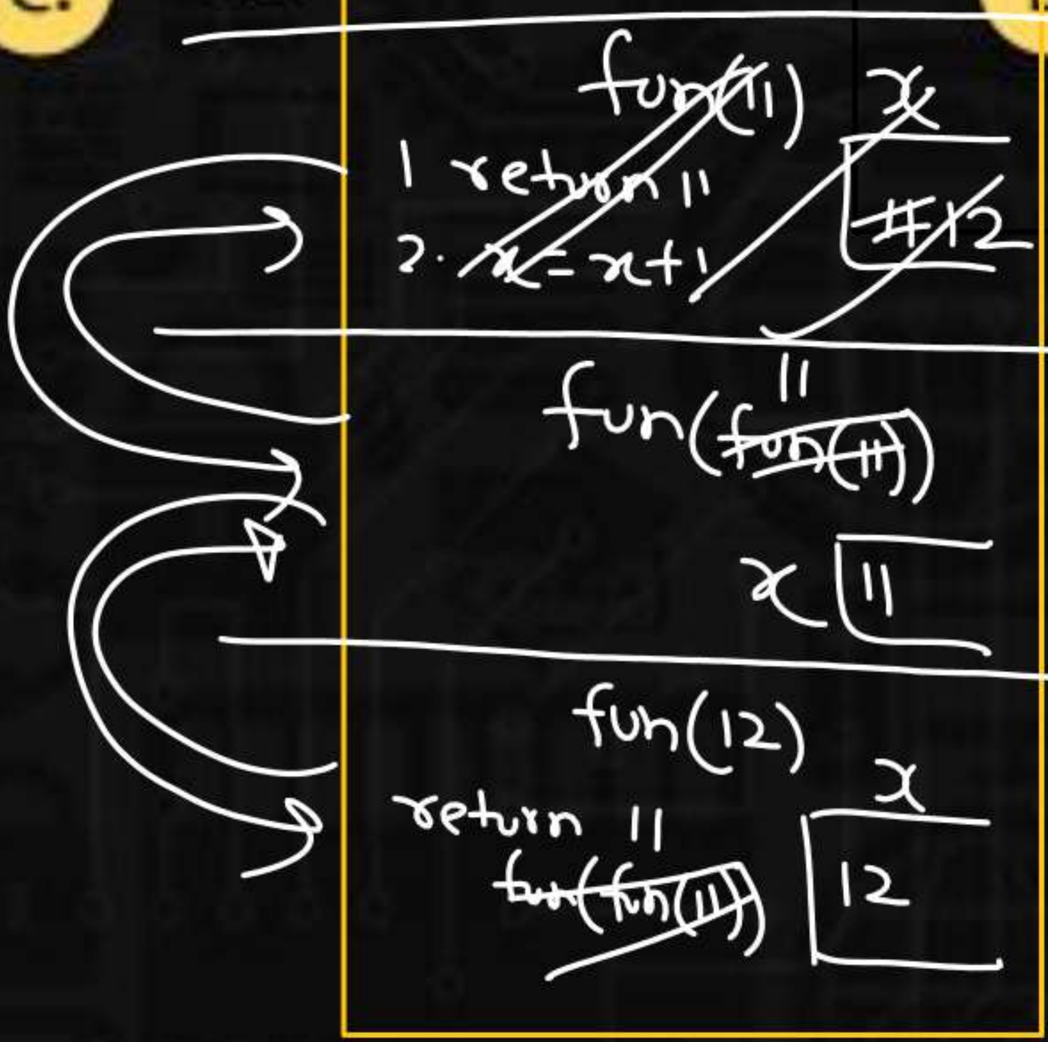
```
int fun(int x)
  {
     if(x%2==0)
     return fun(fun(x-1));
  else
     return(x++);
  }
int main()
  {
                       fun(12)
     printf("%d",f(12));
     [redacted]har();
     return 0;
  }
```

A. 10

B. 11

C. 12

D. None of these



Q.12

```
int fun(int a,int b)
   {
      if(b==0)
      return 0;
      if(b%2==0)
      return fun(a+a,b/2);
      return fun(a+a,b/2) + a;
   }

int main()
   {
      printf("%d",fun(4,3));
      getchar();
      return 0;
   }
```

b is even

A. 12

B. 81

C. 64

D. 8

a b 12
fun(4,3)

a b 8
fun(8,1) + 4

a b 0
fun(16,0) + 8

Q.13

Consider the following C function :

```
int f(int n)
 {
   static int r=0;
   if(n<=0)
   return 1;
   if(n<3)
   {
     r=n;
     return f(n-2) + 2;
   }
   return f(n-1) + r;
 }
```

base

r ~~0~~ 2

what is the value of f(5)

A. 5

B. 7

C. 9

D. 18

f(2) n 2
1 +2
return f(0) + r

f(3) n 3
3+2
return f(2) + r

f(4) n 4
5 +2
return f(3) + r

f(5) n 5
return f(4) + r
7 +2

9

Q.14

Consider the following recursive C function

```
unsigned int foo(unsigned int n, unsigned int r)
 {
   if(n>0)
   return (n%r) + foo(n/r , r);
else
   return 0;
 }
```

output of foo(513,2)

no. of set bits
OR
Count 1s in binary of n

$513 \Rightarrow 2$ 1s

A. 9

B. 8

C. 5

D. 2

Q.14 Consider the following recursive C function

```
unsigned int foo(unsigned int n, unsigned int r)
 {
    if(n>0)
    return (n%r) + foo(n/r , r);
else
    return 0;
 }
```

output of foo(513,2)

A. 9

B. 8

C. 5

D. 2

2

foo(513,2)

1 + foo(256,2)

0 + foo(128,2)

0 + foo(64,2)

0 + foo(32,2)

0 + foo(16,2)

0 + foo(8,2)

0 + foo(4,2)

0 + foo(2,2)

foo(2,2)

0 + foo(1,2)

1 + foo(0,2)

$1289 \% 10$

$\Rightarrow 9$

1000000001 ✓

$513 \% 2 \Rightarrow 1$

$\Rightarrow$

$513 \% 2$

$513/2 = 256$

$$\begin{array}{ccccccccc} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 2^8 & 2^7 & 2^6 & 2^5 & 2^4 & 2^3 & 2^2 & 2^1 & 2^0 \end{array}$$

$\Rightarrow 256$

Q.15 Which of the following statements is/are valid?

A. return a+b;

B. return a,b,c;

C. return (a,b,c);

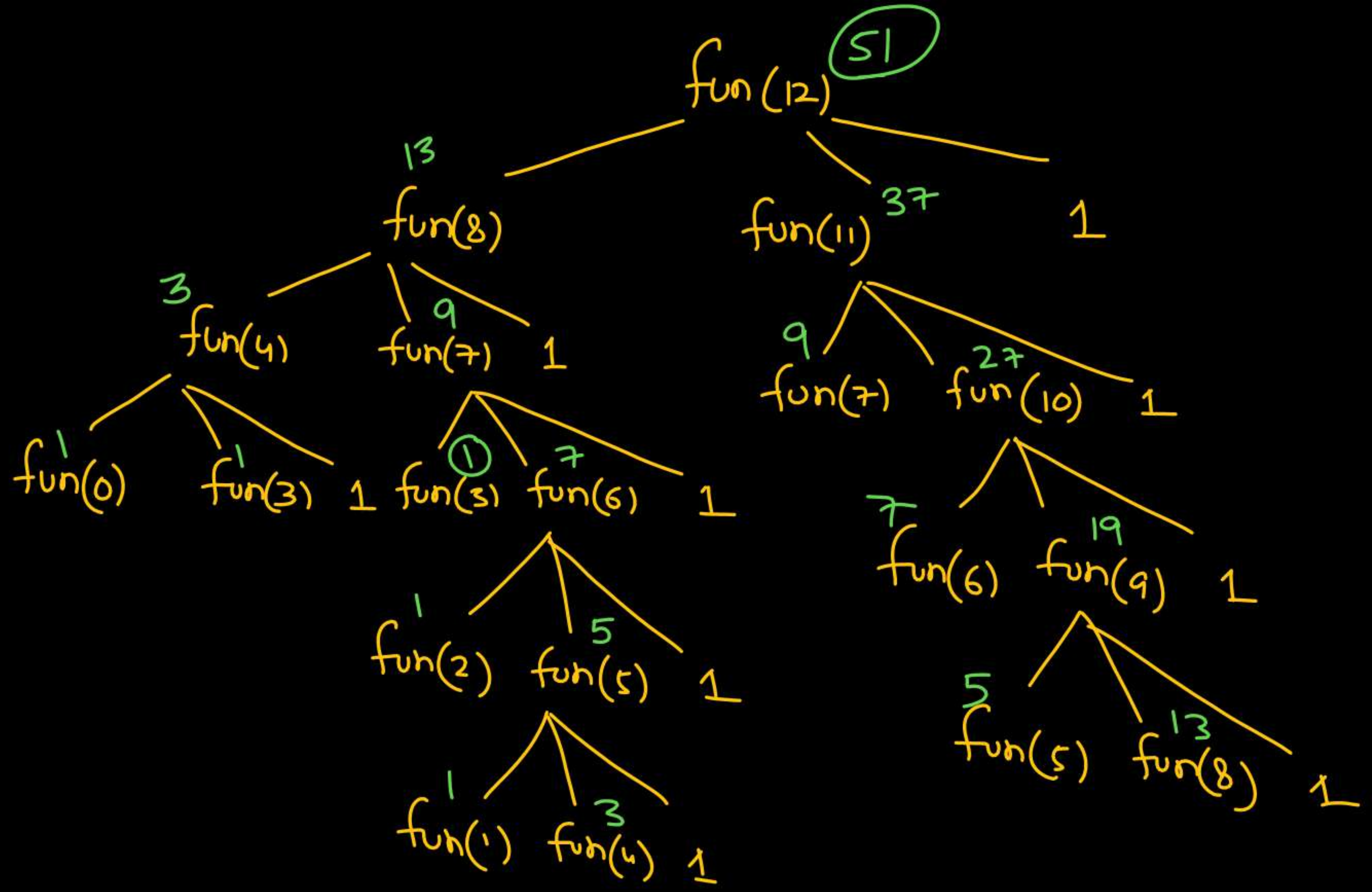
D. All of them

comma operator
sizeof
scoping

Q.16

```
int fun(int x)
 {
   if(x>3)
   return fun(x-4) + fun(x-1) + 1;
   return 1;
 }
```

Find the value returned by fun(12)


51
fun(12)
13
fun(8)
37
fun(11)
1
3
fun(4)
9
fun(7)
1
9
fun(7)
27
fun(10)
1
1
fun(0)
1
fun(3)
1
1
fun(3)
7
fun(6)
1
7
fun(6)
19
fun(9)
1
1
fun(2)
5
fun(5)
1
5
fun(5)
13
fun(8)
1
1
fun(1)
3
fun(4)
1

Q.17

Predict output of following program

```c
#include <stdio.h>
int fun(int n)
{
   if (n == 4)
   return n;
   else return 2*fun(n+1);
}
int main()
{
   printf("%d ", fun(2));
   return 0;
}
```

A. 4

B. 8

C. 16

D. error

16

fun(2)

8

2 × fun(3)

4

2 × fun(4)

Q.18 Consider the following recursive function fun(x, y). What is the value of fun(4, 3)

```
int fun(int x, int y)
{
   if (x == 0)
   return y;
   return fun(x - 1, x + y);
}
```

A. 13

B. 12

C. 9

D. 10

$\text{fun}(\overset{x}{4},\overset{y}{3})$

$\text{fun}(\overset{x}{3},\overset{y}{7})$

$\text{fun}(\overset{x}{2},\overset{y}{10})$

$\text{fun}(1,12)$

$\text{fun}(0,13)$

Q.19

What does the following function do?

```
int fun(int x, int y)
{
  if (y == 0)  return 0;
  return (x + fun(x, y-1));
}
```

A. x + y

B. x + x * y

C. x * y ✓

D. pow(x, y)

$fun(\overset{x}{5}, \overset{y}{3})$

$5 + fun(\overset{x}{5}, \overset{y}{2})$

$5 + fun(\overset{x}{5}, \overset{y}{1})$

$5 + \underline{fun(5, 0)}^{0}$

$\Rightarrow \underline{\underline{3 \times 5}}$

Q.20

What does fun2() do in general?

```
int fun(int x, int y)
{
   if (y == 0)   return 0;
   return (x + fun(x, y-1));
}
```

$x \times y$

```
int fun2(int a, int b)
{
   if (b == 0) return 1;
   return fun(a, fun2(a, b-1));
}
```

A. x * y

B. x + x * y

C. pow(x, y)

D. pow(y, x)

$$fun2(a,b) = a \times fun2(a, b-1)$$
$$= a \times a \times fun2(a, b-2)$$
$$= a^3 \times fun2(a, b-3)$$
$$\vdots$$
$$a^b$$

Q.21

Output of following program?

```
#include<stdio.h>
void print(int n){
    if (n > 4000)
    return;
    printf("%d ", n);
    print(2*n);
    printf("%d ", n);
}
int main()
{
    print(1000);
    [redacted]
    return 0;
}
```

A. 1000 2000 4000

B. 1000 2000 4000 4000 2000 1000

C. 1000 2000 4000 2000 1000

D. 1000 2000 2000 1000

Q.22 What does the following function do?

```
int fun(unsigned int n)
{
  if (n == 0 || n == 1)
    return n;

  if (n%3 != 0)
    return 0;
  return fun(n/3);
}
```

n	o/p
5	0
3	1
6	0

fun(3) → fun(1) → 1

fun(6) → fun(2) → 0

A. ~~It returns 1 when n is a multiple of 3, otherwise returns 0~~

B. It returns 1 when n is a power of 3, otherwise returns 0

C. ~~It returns 0 when n is a multiple of 3, otherwise returns 1~~

D. ~~It returns 0 when n is a power of 3, otherwise returns 1~~

Q.23 Predict the output of following program

```c
#include <stdio.h>
int f(int n)
{
  if(n <= 1)
    return 1;
  if(n%2 == 0)
    return f(n/2);
  return f(n/2) + f(n/2+1);
}
int main()
{
  printf("%d", f(11));
  return 0;
}
```

A. Stack Overflow

B. 3

C. 4

D. 5

Q.24

Consider the following C function:

```
int f(int n)
{
  static int i = 1;
  if (n >= 5)
    return n;
  n = n+i;
  i++;
  return f(n);
}
```

The value returned by f(1) is

A. 5

B. 6

C. 7

D. 8

Q.25 Consider the following C function.

```
int fun (int n)
{
 int x=1, k;
 if (n==1) return x;
 for (k=1; k<n; ++k)
 x = x + fun(k) * fun(n - k);
 return x;
}
```

The return value of fun(5) is ______.

A. 0

B. 26

C. 51

D. 71

Q.26 Consider the following recursive C function. If get(6) function is being called in main() then how many times will the get() function be invoked before returning to the main()?

```
void get (int n)
{
  if (n < 1) return;
  get(n-1);
  get(n-3);
  printf("%d", n);
}
```

A. 15

B. 25

C. 35

D. 45

Q.27

What will be the output of the following C program?

```
void count(int n)
{
  static int d = 1;
  printf("%d ", n);
  printf("%d ", d);
  d++;
  if(n > 1) count(n-1);
  printf("%d ", d);
}
int main()
{
  count(3);
}
```

A. 3 1 2 2 1 3 4 4 4

B. 3 1 2 1 1 1 2 2 2

C. 3 1 2 2 1 3 4

D. 3 1 2 1 1 1 2

Q.28

What will be the output of the C program?

```
#include<stdio.h>
int main()
{
  function();
  return 0;
}
void function()
{
  printf("Function in C is awesome");
}
```

A. Function in C is awesome

B. no output

C. Runtime error

D. Compilation error

Q.29

What will be the output of the C program?

```
#include<stdio.h>
int main()
{
  main();
  return 0;
}
```

A. Runtime error

B. Compilation error

C. 0

D. None of these


THANK YOU GW
SOLDIERS !